

PERFORMANCE / VERIFICATION

Performance/Verification is run via the REASON pulldown menu of the Calibration selection of the Service Application.

For *system performance tests*, *test methods*, and *performance specifications*, refer to the [Performance Specifications](#) section of this manual.

Manual Performance is run the same way as auto, except the REASON menu is different to allow only certain calibration/performance routines. Refer to the table below for the Manual Calibration/Performance pulldown menu. (Performance routines are in bold).

Reason Pull Down Menu – MANUAL CALIBRATION

REASON	DESCRIPTION OF ACTION PERFORMED
ManPerform_Lateral_Centering	
ManFind_Centering	
ManFind_DAC	
MandFind_Air	
MandFind MA	
ManFind_Flatts	
ManFind_FlattsFF	
ManFind_Level	
ManFind_LevelFF	
ManFind_Level2	
ManFind_Level2FF	
ManFind_PilotAir	
ManPerform_MTF	
Clear_BadDetsList	

How to Run Manual Performance / Calibration

1. After the reason is selected, a text editor appears, containing the calibration/performance routine text of the Reason that was selected. Find and edit the appropriate text parameters for the calibration or performance routine you selected. To edit a routine, highlight the text value (mA, voltage, etc.) with the mouse pointer. (Click on the text, hold and drag across the text of choice until the text displays in reverse video.) Refer to Figure 110.
2. After the text is highlighted, type in the new value. For example, select 130kV and type in 120kV. After the text is edited, save the file and close the text editor window. When closing the editor, you will be prompted with a information about the steps you should perform. Refer to Figure 111. (In this example, the ManFind_Centering Reason was chosen.) Click on OK to continue.
3. From the scanning screen, click on the Proceed to Scan button, then the Start Scan button to continue with the Performance routine.

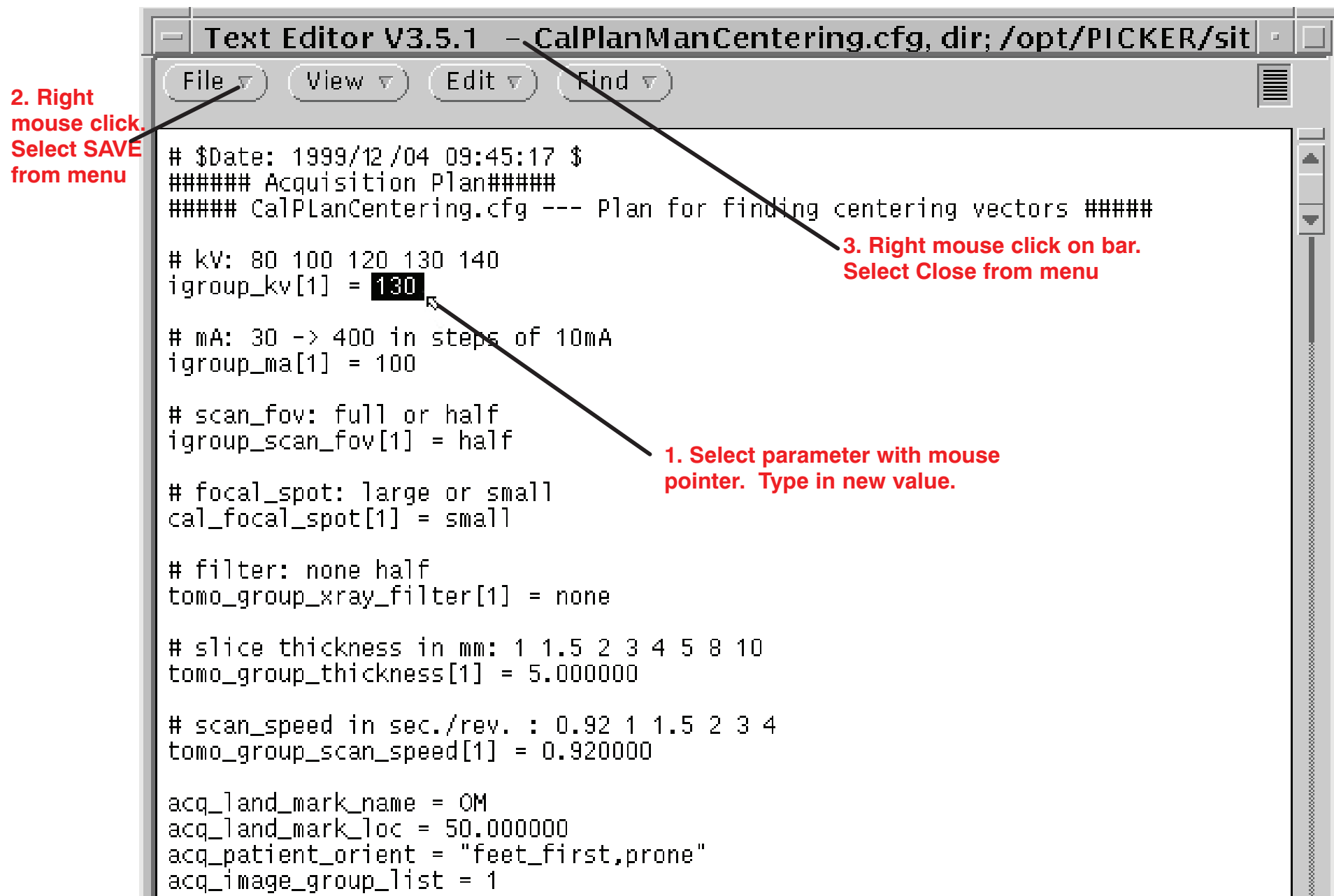


FIGURE 110 MANUAL PERFORMANCE / CALIBRATION TEXT EDITOR

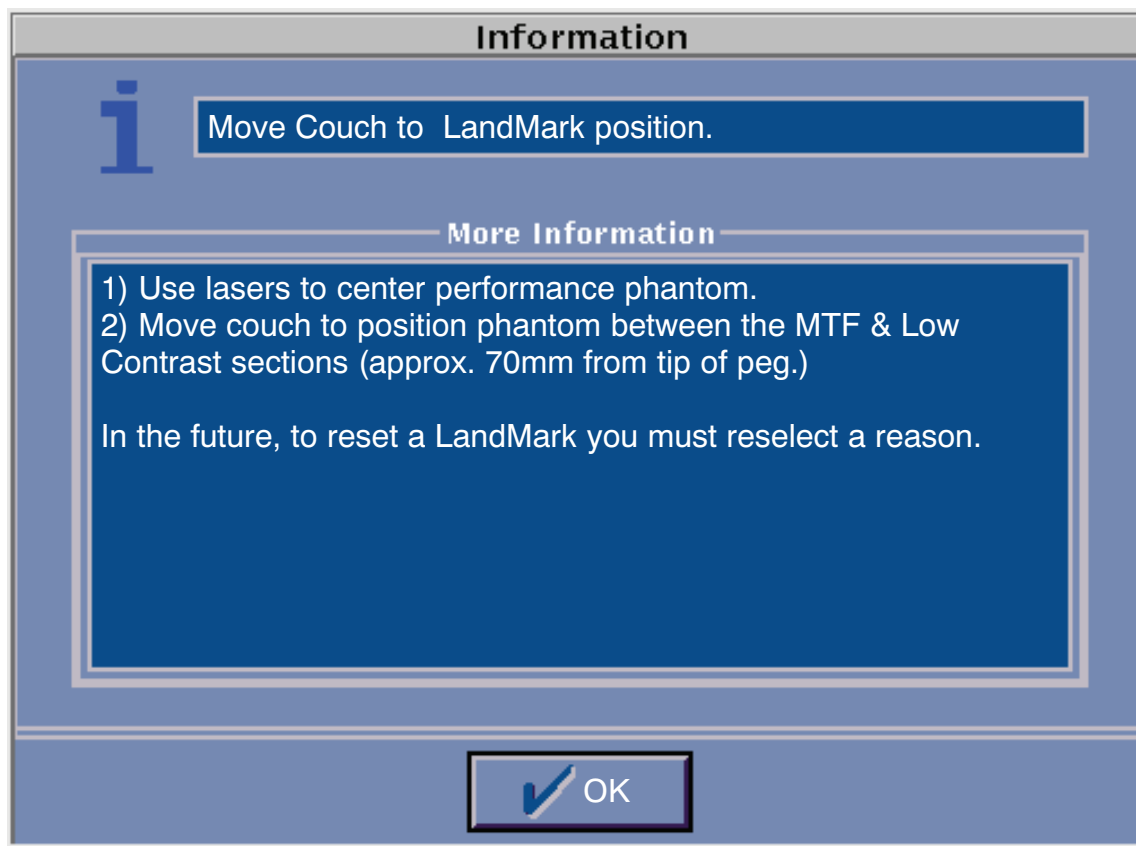


FIGURE 111 MANUAL PERFORMANCE / CALIBRATION INFORMATION WINDOW

PERFORMANCE SPECIFICATIONS

INTRODUCTION

This document provides system performance tests, test methods, and performance specifications for the AcQSim CT CT System. The test methods assume familiarity with the operation of the AcQSim CT System. A AcQSim CT System satisfying these performance specifications is a necessary but not sufficient condition for acceptable system performance.

TEST PHANTOM

The test phantom used for the majority of these tests is the Philips Medical Systems Performance Phantom. This phantom is supplied with the AcQSim CT System. This phantom is specially designed to evaluate the image performance of Philips Medical Systems CT systems and consists of the six separate scanable sections as described below. The standard phantom landmark for many of the tests is the crease between the fourth and fifth sections.

1. A 12 inch diameter uniform water section is used for full field uniformity and noise.
2. A 8 inch diameter uniform water section is used for half field uniformity and noise.
3. A 6 inch diameter uniform water section is used for low contrast measurements.
4. A 6 inch diameter low contrast section contains a low contrast plastic insert (.35% contrast compared to water) drilled with 7 rows of four holes (6mm, 5mm, 4mm, 3mm, 2.5mm, 2mm, and 1.5mm). This plastic section is surrounded by water and also contains a large acrylic peg and large polyethylene peg for measuring the linearity of the CT numbers.
5. A 6 inch diameter slice thickness / MTF section is a plastic section which contains two foil strip ramps or wedges whose angle is 14 degrees from the phantom's central axis. These ramps are used for measuring slice thickness. Also imbedded in this plastic section is a .05mm diameter tungsten wire used for line spread function measurements leading to MTF results.
6. The last section is a removable 1 inch diameter plastic cylinder which contains a central pin which is useful for phantom centering and alignment with the system's central axis of rotation.

DESIGN

The following list contains descriptions of the AcQSim CT System's design and notes differences from the Ultra Z product (version 1.0) that are relevant for this T95 document:

Detectors and Detector Modules:

The gantry ring has 2400 individual CdWO₄ detector elements. These detector elements are divided into 120 detector modules.

Each Module contains two (10 element) crystal arrays (29mm long and 2.26mm thick);

Each Crystal array (21.92mm wide) has 10 defined channels ;

Each Channel has a 1.58mm detector space;

Detector (center-to-center) distance is 2.17mm.

SDD:

The radial distance (863mm) of the detector hoop is maintained from the distance established by the Ultra Z. The SCD (focal spot to imaging isocenter) is extended with respect to the Ultra Z from 635mm to 651mm for Rhino 4/5/6.5 X-ray tubes.

Dose Efficiency:

Assuming negligible spacing between modules and crystals:

Detector Dose Efficiency_{AcQSim CT} = Q.D.E.*Geometric Efficiency

QDE~100%

Geometric Efficiency= Detector space/(Module space + Intermodule dead space)

Geometric Efficiency~ $15.8 / (21.9 + (2 * \pi * 13 / 240)) \sim 15.8 / 22.3 = 70\%$

Geometric Dose Efficiency: Since the tube has been moved out to accommodate the larger bore, the dose to the patient will be reduced (with respect to the Ultra Z).

Sampling:

Half Field scans are not supported by the AcQSim CT system.

In Full-field, 3 samples per detector angle of rotation (.15 degrees) for 512 detectors.

FOV

The Full Field field of view is extended from 48 cm (Ultra Z) to 60 cm.

PREREQUISITES

1. The factory laser alignment procedure has been successfully completed.
2. Couch calibration has been completed and tilt position is correct.
3. A complete calibration, i.e. changed tube calibration, has run successfully. This complete calibration means that electronic centering, DAC, air calibration, MA curves, pilot cals, spectrum and scaling constants have been generated. It also means that mechanical alignment of the x-ray tube, both lateral and axial have been generated.
4. This procedure is intended to be used only on systems running with AcQSim CT 1.0 Software (or higher) system software and AcQSim CT 1.0 (or higher) diagnostic software.
5. All diagnostic programs and software tools referenced within this document, or their functional equivalent, must be resident in the software (i.e. cna baco, cna resolver, cna water, mtf, kvhvl, lowcon, slice thickness, and uniformity).
6. AutoVoice and auto-archive must be "OFF" for all tests. Data Save must be off unless otherwise specified.
7. Prior to beginning this performance test, it must be verified that an open circuit (>10 Mohm) exists between the ground block (DC Common) and the AC Common wire. This can be performed simply by removing the green/yellow striped wire from the ground block located on the cardfile and measuring with an ohmmeter (20M range).
8. Ensure that the face of the phantoms used during these tests are positioned parallel to the central xray of the fan beam. Run Center_Phantom to insure correct positioning of the Philips Medical Systems performance test phantom.
9. The number and location of bad detectors is within the system specification for bad detectors. See below.

System Spec for Bad Detectors

- a. There can be no more than 16 bad detectors per system / else replace the module with the most bad detectors.
- b. There can be no more than 3 bad channels per detector module / else replace the module.
- c. There can be no more than 2 adjacent bad detectors, either in a module or in adjacent modules / else replace the module.
- d. No defective channel can adversely effect other channels in a quadrature manner / else replace the defective component causing the adverse effect.
- e. Any module with a defective ARDAS channel must be replaced.

**** VERIFICATION ****

1. A permanent record (as a logfile or data sheet) of all system performance data of the scanner must be stored via digital archive or hard copy.

TEST 1 – LASER ALIGNMENT TEST – FIRST PILOT –AUTO POSITIONING

Purpose: To verify that the alignment laser beams align with the transaxial x-ray beam. Also to verify the pilot scan operation and the couch positioning are correct. Test Parameters and Method: Manual

1. Enter the Clinical Scanning or Console mode of operation.
2. Select the Adult Brain protocol.
3. Center and align the 6 inch water section of the Philips Medical Systems Performance Phantom on the CT System couch and gantry using the patient alignment lasers. The laser lights should bisect the phantom.
4. Perform a vertical pilot scan of the phantom with a pilot range of +150 to –150 mm. After the pilot image is displayed, note that the image of the phantom is centered in the pilot scanned area. Adjust the window setting and zoom the image to optimally view the phantom.
5. In the pilot scan find the dark rectangular portion in the last section of the phantom that is a slot for hanging the centering rod or additional phantoms. Position the plan line at the exact bottom of this rectangular region which is between the endplate and the slice thickness/mtf section.
6. Select a Cine scan type and record the co-ordinate location of the plan line under Extent: _____.
7. Start the Cine scan and observe the resulting axial image. Set the level to 100 and the window to 200. The table is positioned properly if the two parallel bars (slice thickness wedges) and the oval portion (bisecting the upper bar) are visible in the image. Note the final couch position recorded under scan status: _____.
8. Turn on the lasers and verify that the beams fall directly on the seam between the mtf/slice thickness section and the endplate. If necessary, adjust laser positions on the x-ray frame and repeat the laser alignment test if one or more of the lasers do not fall on the seam.
9. Record all test results below or on a system performance data sheet.

Results and Specification:

Pilot Scan Operation [pilot image centered]	(ok)
Pre Cine Scan Plan Location [extent position]	
Post Scan Couch Location (within .5mm of previous line)	
Couch Positioning (Parallel bars and oval visible in image)	(ok)
Laser / X-Ray Beam Alignment (Laser line falls on phantom seam correctly)	(+/- 0.5mm)

TEST 2 – CHECK FOR BAD DETECTORS

Tests and Purposes: **BadDetsBaco**: Determine each detector's background noise and offset level and compare against specification.

BadDetsWater: Determine each detector's noise level on a subtracted 12 inch water phantom scan and compare against specification.

Insure that there is no air bubble in the phantom as this could cause the test to fail.

BadDetsAir: Determine each detectors dc response level for an air scan and compare against specification.

BadDetsPEG: Determine each detector's pin area on a subtracted peg from pin scan and compare against specification.

Test Parameters: (All set automatically by the service application software test method)

KV	100	130	130	130
Filter	0	0	0	0
Focal Spot	Small	Small	Small	Small
Scan Field	Full	Full	Full	Full
Slice Thickness	8mm	8mm	8mm	5mm
MA	30	130	90	70
Frame Speed	2	4	4	4
Matrix_Y	512	512	512	512
Recon_type	Normal	Over	Over	Over
Recon_Field	na	600	600	600
Couch	na	+106mm #1 +108mm #2	–160mm	–39mm for scan #1 –60mm for scan #2
Conditions				
Acq_chspd	0	0	0	0
Samples/unit	3	3	3	3
Src_active_dets	648	648	648	648
Calcium_correction	Off	Off	Off	Off
Scan_Start_Degrees	Variable	Variable	Variable	Variable
Off set Frequency	12 KHz	na	na	na
Scan Type	na	Scan	Scan	Scan
Command_type	Scan	Scan	Scan	Scan

Test Method:

- Service Application
- Calibrate WorkStep
- Center_Phantom
- Find_BadDets Reason

- BadDetsBaco
- BadDetsWater
- BadDetsAir
- BadDetsPeg

Results and Specification:

BadDetsBaco	BadDetsWater
Ave offset: _____ (ref. $e-02v$)	Water Ave. Noise: _____ (ref. $e-04v$)
Max offset: _____ ($-3.50e-02 v$)	Water Max Noise: _____ ($6.00e-04 v$)
Min offset: _____ ($-2.50e-02 v$)	Water Min Noise: _____ ($\geq 1.70e-04 v$)
Ave noise: _____ (ref. $e-05v$)	
Max noise: _____ ($7.00e-05 v$)	
Min noise: _____ ($\geq 1.5.e-05 v$)	

BadDetsAir	BadDetsPeg
Air Ave. Response: _____ (ref. $+00v$)	Peg Area Ave. : _____ (ref. $e +04$)
Air Max Response : _____ ($-6.00e+00 v$)	Peg Area Max. : _____ (ref. $e +04$)
Air Min Response: _____ ($-2.00e+00 v$)	Peg Area Min. : _____ (ref. $e +04$)

All detector Peg Area Values must be within $\pm 12\%$ of the system average, else place the offending detector on the bad list.
 Specification: All BadDets tests must run to completion without errors.

TEST 3 – PERFORM CNA (COHERENT NOISE ANALYSIS)

Tests and Purposes: **Perform CNA****Baco:** Perform spectrum analysis on detector fans for a background scan and verify that background noise is within specifications.

Perform CNA**Water:** Perform Spectrum Analysis on detector fans of two subtracted 8 inch water scans and verify that the noise amplitudes are within specification.

Perform CNA**Resolver:** Perform spectrum analysis of detector fan data of a digital check scan and verify that the peak to peak arc–sec amplitudes are within specification.

Key Test Parameters: (All set automatically by software method)

KV	130	130	130
Filter	0	0	0
Focal Spot	Large	Large	Large
Slice Thickness	8mm	8mm	8mm
MA	120	150	120
Scan Field	Full	Full	Full
Frame Speed	2	2	2
Matrix_Y	512	512	512
Recon_type	Normal	Over	Norm
Conditions			
Samples/unit	3	3	3
Src_active_dets	648	648	648
Acq_chspd	3	3	3
Command_type	Scan	Scan	Scan
Scan_Start_Degrees	Variable	0 degrees	Variable
Scan Digital Check Freq.	na	na	2MHz
Background Freq	12kHz	na	na

Test Method:

- Service Application
- Calibrate WorkStep
- Perform _CNA Reason
 - CnaBaco
 - CnaWater
 - CnaResolver

Test Results and Specifications:

CnaBaco		
Frequency	Measured Result	Specification
Any one $\geq$ 30 hz	Highest Value =	$\leq 4.00\text{e-}06$ vp
White Noise (Total)		$\leq 5.0\text{e-}05$ vp

CnaWater		
Frequency Range (hz)	Highest Measured Value	Specification (vp)
10 – 100hz		$\leq 3.5\text{e-}04$ vp
100 – 399hz		$\leq 9.0\text{e-}04$ vp
400 – 650hz		$\leq 3.7\text{e-}04$ vp
> 650hz		$\leq 1.4\text{e-}04$ vp
White Noise (Total)		$\leq 5.0\text{e-}03$ vp

CnaResolver		
Frequency	Measured Result	Specification (arc sec)
450 cycles/rev		< 3.35 arc–sec p–p
Any frequency other than 450cycles/rev	Highest Value =	≤ 1.67 arc–sec p–p
White Noise (Total)		< 2.0e+01 arc–sec p–p

TEST 4 – BEAM QUALITY – KVHVL

Purpose: Measure the half value layer of the system and verify that it meets specification.

Test Parameters: : (All set automatically by the service application software test method:

Compensator	None
KV	130
Filter	0
Focal Spot	Large
Scan Field	Full
Slice Thickness	2mm
MA	130
Frame Speed	4
Matrix_Y	512
Recon_type	Over
Recon_Field	240
Algorithm	Smooth
Couch	Move couch –159mm relative to phantom 0 for scan #1 with kvhvl fixture Move couch –161mm relative to phantom 0 for scan #2 w/o kvhvl fixture
Conditions	
Acq_chspd	0
Scan Type	Scan
Samples/unit	3
Src_active_dets	648
Calcium_correction	Off
Command_type	Scan
Scan_Start_Degrees	0

Test Method:

- Service Application
- Calibrate WorkStep
- Perform_KVHVL Reason

Test Results and Specifications:

HVL (@130 KV)	_____ (3.7 to 4.5 mm of aluminum)
HVL (@100 KV)	_____ extrapolated from 130 KV measurement (2.8 to 3.7 mm)
Calculated KVp	_____ (For reference only; approx. 130KVp)

TEST 5 – MTF

Purpose: Determine the Modulation Transfer Function (MTF) by scanning a thin wire and verifying that the system meets specification.

Test Parameters: : All set automatically by the service application software test method:

3 MTF Measurements are made:		#1	#2	#3
KV	130			
Filter	0			
Focal Spot		Small	Large	Small
Scan Field	Full			
Slice Thickness	8mm			
MA	130			
Frame Speed		4	4	1
Matrix_Y	512			
Scan Type	Scan			
Recon_type	NORM			
Recon_Field	60mm			
Algorithm	MTF			
Couch	Move couch to –9mm relative to phantom 0 for scan #1, #2, #3.			
Conditions				
3 MTF Measurements are made:				
Command_type	Scan			
Scan_Start_Degrees	Variable			
Samples/unit	3			
Src_active_dets	648			
Calcium_correction	Off			
Acq_chspd	0			

Test Method:

- Service Application
- Calibrate WorkStep
- Perform_MTF Reason

Test Results and Specifications:

	Scan #1 (Small Spot)	Scan #2 (Large Spot)	Scan #3 (1 sec; Small Spot)
50% modulation	_____ (7.0 lp/cm)	_____ (6.8 lp/cm)	_____ (7.0 lp/cm)
10% modulation	_____ (12.0 lp/cm)	_____ (11.5 lp/cm)	_____ (11.5 lp/cm)
5% modulation	_____ (13.0 lp/cm)	_____ (12.6 lp/cm)	_____ (12.6 lp/cm)

TEST 6 – SLICE THICKNESS

Purpose: Measure various slice thicknesses and verify that the system meets the specification.

Test Parameters: All set automatically by the service application software test method.

KV	130	130	130	130
Filter	0	0	0	0
Focal Spot	Large	Large	Large	Large
Scan Field	Full	Full	Full	Full
Slice Thickness	8mm	5mm	3mm	2mm
MA	100	100	100	100
Frame Speed	4	4	4	4
Matrix_Y	512	512	512	512
Recon_type	Norm	Norm	Norm	Norm
Recon_Field	60mm	60mm	60mm	60mm
Conditions:				
Acq_chspd	0	0	0	0
Scan Type	Scan	Scan	Scan	Scan
Samples/unit	3	3	3	3
Src_active_det	648	648	648	648
Calcium_correction	Off	Off	Off	Off
Command_type	Scan	Scan	Scan	Scan
Scan_Start_Degrees	Variable	Variable	Variable	Variable

Test Method:

Service Application
Calibration Workstep
Perform_

Test Results and Specifications:

Nominal Slice Thickness	Results	Specification
8mm		7.2 to 8.8mm
5mm		4.5 to 5.5mm
3mm		2.5 to 3.5mm
2mm		1.5 to 2.5mm

TEST 7 – WATER UNIFORMITY AND NOISE

KV	80	100	120	130	140
Scan Field	Full	Full	Full	Full	Full
Filter	0	0	0	0	0
Focal Spot	Large	Large	Large	Large	Large
Algorithm	Soft Tissue	Soft Tissue	Soft Tissue	Soft Tissue	Soft Tissue
MA	130	130	130	130	130
Frame Speed	4	4	4	4	4
Slice Thickness	8mm	8mm	8mm	8mm	8mm
Recon Field	480 mm	480 mm	480 mm	480 mm	480 mm

Test Method:

- Service Applications
- Calibrate WorkStep
- Perform_Uniformity: Measurements made on center of fov, at 1/2 radius for peripherals. Measurements using a 2000mm² square ROL.

Results and Specifications:

FF Uniformity and Noise (Filter 0)	80KV	100KV	120KV	130KV	140KV
Center Value (0) = (0 +/- 2.0)					
Highest Perif Value (0) =(C.V. +/-3.0)					
Full Field Noise Spec (x10)	7.5	7	7	7	7
(Max. Std. Dev. in HU)					

TEST 8 – LOW CONTRAST RESOLUTION

Purpose: To measure the low contrast resolution of the system under certain clinical parameters and determine if the low contrast resolution is within specification.

Test Parameters: (All set automatically by the service application software test method)

KV	130
Filter	0
Focal Spot	Small
Scan Field	Full
Slice Thickness:	8mm
MA	160
Frame Speed	4
Matrix_Y	512
Recon Type	Over
Recon Field	250
Algorithm	Soft Tissue
Couch	Move couch +16mm relative to phantom 0 for scan #1 Move couch +40mm relative to phantom 0 for scan #2
Conditions:	
Acq_chsp	0
Scan Type	Scan
Samples/Unit	3
Src_active_dets	648
Calcium_correction	Off
Command_type	Scan
Scan_start_degrees	Variable

Test Method:

- Service Applications
- Calibrate WorkStep
- Perform_LowCon_Reason

Test Results and Specifications:

Low Contrast Measurement		0.25 to 0.55%
Noise in Low Con Section		≤ 1.55 HU std. dev. of water
Hole Size at 0.35% contrast		≤ 3.5 mm
CT Number of Water		0 $\pm$ 5 HU
CT Number of Acrylic (peg1)		120 $\pm$ 10 HU
CT Number of Polyethylene (peg2)		-96 $\pm$ 5 HU

TEST 9 – MULTI–SLICE STUDY CYCLE TIME

Purpose: Measure the study cycle time for a common time consuming multi–slice study and verify that the system meets the timing specification.

Test Parameters: This is a manual test.

Test Object	8" Water Phantom
Protocol	Abdomen
Algorithm	Soft Tissue
Scan Time	1.5 sec
mA	120
KV	130
Slice Thick	8mm
Scan Angle	normal
Couch Increment	8mm
Data save	Off

Test Method:

1. Verify that the starting tube heat is less than 30%.
2. Plan an axial scan sequence of 30 axial images and Plan / Edit any of the specified scan parameters of the protocol to match the above specified parameters.
3. Total study time is measured from x–ray on of the first scan to image preview complete for the 30 th scan.

Results and Specification:

Time from first to last exposure =	_____ (≤ 85sec.)
Total time from first exposure to last image review =	_____ (≤ 90sec.)

TEST 10 – SPIRAL SCAN STUDY CYCLE TIME

Purpose: Measure the spiral study cycle time for a common time consuming spiral study and verify that the system meets the timing specification.

Test Parameters: This is a manual test.

Test Object	8" Water Phantom
Protocol	Abdomen
Algorithm	Soft Tissue
Scan Time	1.5 sec per revolution
mA	120
KV	130
Slice Thick	8 mm
Index	8 mm
Pitch	1
Interpolation	Standard
PVC	Off
Data save	Off

Test Method:

1. Verify that the starting tube heat is less than 30%.
2. Plan a spiral scan sequence of 29 axial images and Plan/Edit any of the specified scan parameters of the protocol to match the above specified parameters.
3. Total study time is measured from x-ray on of the first scan to image preview complete for the 29th scan.

Use either the system's line profile and distance measurement utilities or the background to disappearing ramp method. See the CTDI document for a detailed description of this latter method].

Time from first to last exposure =	_____ (≤ 48sec.)
Total time from first exposure to last image review =	_____ (≤ 80sec.)

TEST 11 – BACKUP TIMER (OPTIONAL)

PURPOSE

This procedure defines the methods to be followed to verify the scan time, pilot time and x-ray back-up timer are within design parameters.

EQUIPMENT REQUIRED

Standard Test Technician Tool Box.

Gould RS 3200 strip chart recorder.



CAUTION

WHEN HANDLING PRINTED CIRCUIT BOARDS OR ANY ESD SENSITIVE DEVICES
USE APPROPRIATE ESD GROUNDING PRECAUTIONS.

NOTE

Unless otherwise specified the system should be powered up and the computer booted up.

NOTE

Ensure the tube has been properly warmed up prior to performing sections that produce x-rays.

PROCEDURE

1. Raise the patient support to a height that will allow the carbon top to pass through the center of the scanner before starting the test.
2. Put the GPUC on an extender card.
3. Attach a chip clip to U13.



 CAUTION

THE LENGTH OF THE CHIP CLIP CAUSES THE PINS TO BE VERY CLOSE TO THE GANTRY SIDE COVER. CARE MUST BE TAKEN THAT THE CHIP CLIP LEADS DO NOT SHORT AGAINST THE GANTRY SIDE COVER.

4. On the Gould chart recorder set the D.C. amplifier (channel 1) as follows:

Full Scale Voltage	.05
--------------------	-----

Sensitivity x100

Filter OFF

5. Connect both leads of channel 1 of the chart recorder to TP1 (D.C. common on the GPUC card).

6. Start the chart recorder at a speed of 5 mm per sec, center the ink pen on the graph, and stop the chart recorder.

7. Connect the signal lead of channel 1 to the x-ray start/ stop pulse (XRSTSP) located at U13 pin 3 on the GPUC board. Connect the negative lead to TP1.

8. Modify **CENTER CAL** protocol in the CLINICAL mode as follows:

SCAN TIME: 4 FOV: **LARGE** SCAN ANGLE: **OVER**

9. Perform a scan. Prior to clicking the 'start' button on the screen, start the chart recorder at a speed of 100mm/sec.

10. At the completion of the scan, stop the recorder and compute the time between the two "XRSTSP" pulse transitions. Record this time on data sheet.

11. Modify **CENTER CAL** protocol in the CLINICAL mode as follows:

SCAN TIME: 2 FOV: **LARGE** SCAN ANGLE: **NORM**

12. Perform a scan. Prior to clicking the 'start' button on the screen, start the chart recorder at a speed of 100mm/sec.

13. At the completion of the scan, stop the recorder and compute the time between the two "XRSTSP" pulse transitions. Record this time on data sheet.

14. Modify **CENTER CAL** protocol in the CLINICAL mode to perform a pilot scan. Position the patient support top to allow for a travel of 256mm on each side of the landmark. Change the PILOT RANGE to:
START: -256 END: +256
15. Perform a pilot scan. Prior to clicking the 'start' button on the screen, start the chart recorder at a speed of 100mm/sec.
16. At the completion of the pilot scan, stop the recorder and compute the time between the two "XRSTSP" pulse transitions. Record this time on data sheet.
17. Move the carbon top to its outermost position. Ensure there are no obstructions in the scan field.
18. Switch from CLINICAL to DIAGNOSTIC MODE.
19. On the Gould chart recorder set the D.C. amplifier (channel 1) as follows:
Full Scale Voltage .25
Sensitivity x100
Filter OFF
20. Connect both leads of channel 1 of the chart recorder to DC common.
21. Start the chart recorder at a speed of 5 mm per sec, center the ink pen on the graph, and stop the chart recorder.
22. Remove the J1 connector from VFSC board 5 (For AcQSim CT use VFSC board 7). Insert an extender pin in J1 pin 4 and J1 pin 1. Connect the chart recorder positive lead to pin 1 and the common lead to pin 4.
23. From an Xterm window change the directory by typing: cd/opt/PICKER/picr/service_util/diagnostics/pq2000p/olevel/bin
24. At the prompt type: **backup_timer**

25. Execute macro **@povt4**. Start the chart recorder at a speed of 100mm/sec when instructed to do so. This macro will generate an exposure to verify the back-up timer.
26. Stop the chart recorder once the x-ray over time error occurs.
27. Compute the actual exposure time from the strip chart read-out. Record this time on data sheet.
28. Execute macro **@povt2**. Start the chart recorder at a speed of 100mm/sec when instructed to do so. This macro will generate an exposure to verify the back-up timer.
29. Stop the chart recorder once the x-ray over time error occurs.
30. Compute the actual exposure time from the strip chart read-out. Record this time on data sheet.
31. Return the system to the original configuration.

SYSTEM PERFORMANCE DATA SHEET

Date: _____

System I.D.: _____

TEST 1: COUCH/LASER POSITION

Pilot Scan Operation [pilot image centered]	_____ (ok)
Pre Cine Scan Plan Location [extent position]	_____
Post Scan Couch Location [within .5mm of previous line]	_____
Couch Positioning [Parallel bars and oval visible in image]	_____ (ok)
Laser / X-Ray Beam Alignment [Laser line falls on phantom seam correctly]	_____ (+/- .5mm)

TEST 2: CHECK FOR BAD DETECTORS

BadDetsBaco	Ave offset	_____ (ref. e-02v)
	Max offset	_____ (-3.50e-02 v)
	Min offset	_____ (-2.50e-02 v)
	Ave noise	_____ (ref. e-05v)
	Max noise	_____ (7.00e-05 v)
	Min noise	_____ ($\geq 1.5.e-05$ v)

BadDetsWater	Water Ave. Noise	_____ (ref. e-04 v)
	Water Max Noise	_____ (6.00e-04 v)
	Water Min Noise	_____ ($\geq 1.70e-04$ v)

BadDetsAir	Air Ave. Offset	_____ (ref. +00v)
	Air Max Offset	_____ (-6.00e+00 v)
	Air Min Offset	_____ (-2.00e+00 v)

BadDetsPeg	Peg Area Ave. Offset	_____ (ref. e +04)
	Peg Area Max. Offset	_____ (ref. e +04)
	Peg Area Min. Offset	_____ (ref. e +04)

All detector Peg Area Values must be within +/- 12% of the system average, else place the offending detector on the bad list.
Specification: All BadDets tests must run to completion without errors.

TEST 3: COHERENT NOISE ANALYSIS (CNA)

CnaBaco	Frequency	Measured Result	Specification
	Any one ≥ 30 hz	Highest Value =	$\leq 4.00\text{e-}06$ vp
	White Noise (Total)		$\leq 5.0\text{e-}05$ vp

CnaWater	Frequency Range (hz)	Highest Measured Value	Specification (vp)
	10 – 100hz		$\leq 3.5\text{e-}04$ vp
	100 – 399hz		$\leq 5.0\text{e-}04$ vp
	400 – 650hz		$\leq 3.7\text{e-}04$ vp
	> 650hz		$\leq 1.4\text{e-}04$ vp
	White Noise (Total)		$\leq 5.0\text{e-}03$ vp

CnaResolver	Frequency	Measured Result	Specification (arc sec)
	450 cycles/rev		< 3.35 arc–sec p–p
	Any freq other than 450cycles/rev	Highest Value =	≤ 1.67 arc–sec p–p
	White Noise (Total)		< 2.0e+01 arc–sec p–p

TEST 4: BEAM QUALITY – KVHVL

HVL (@130 KV) _____	(3.7 to 4.5 mm of aluminum)
HVL (@100 KV) _____	extrapolated from 130 KV measurement (2.8 to 3.7 mm)
Calculated KVp _____	For reference Only; approx. 130KV)

TEST 5: MTF

	Scan #1 (Small Spot)	Scan #2 (Large Spot)	Scan #3 (1 sec; Small)
50% modulation	_____ (7.0 lp/cm)	_____ (6.8 lp/cm)	_____ (6.8 lp/cm)
10% modulation	_____ (12.0 lp/cm)	_____ (11.5 lp/cm)	_____ (11.5 lp/cm)
5% modulation	_____ (13.0 lp/cm)	_____ (12.6 lp/cm)	_____ (12.6 lp/cm)

TEST 6: SLICE THICKNESS

Nominal Slice Thickness	Results	Specification
8mm		7.2 to 8.8mm
5mm		4.5 to 5.5mm
3mm		2.5 to 3.5mm
2mm		1.5 to 2.5mm

TEST 7: WATER UNIFORMITY AND NOISE

FF Uniformity and Noise (Filter 0)	80KV	100KV	120KV	130KV	140KV
Center Value (0) = (0 +/- 2.0)					
Highest Perif Value (0) =(C.V. +/- 3.0)					
FF (0) Noise Spec (x10)	7.5	7.00	7.00	7.0	7.00
(Max. Std. Dev. in HU)					

TEST 8: LOW CONTRAST RESOLUTION

Low Contrast Measurement	_____ (0.25 to 0.55%)
Noise in Low Con Section	_____ (≤ 1.5 HU std. dev. of water)
Hole Size at 0.35% contrast	_____ (≤ 3.5 mm)
CT Number of Water	_____ (0 $\pm$ 5 HU)
CT Number of Acrylic (peg 1)	_____ (120 $\pm$ 10 HU)
CT Number of Polyethylene (peg 2)	_____ (-96 ± 5 HU)

TEST 9: MULTI-SLICE STUDY CYCLE TIME (30 Axial Slice Sequence)

Time from first to last exposure =	_____ (≤ 85 sec.)
Total time from first exposure to last image review =	_____ (≤ 90 sec.)

TEST 10: SPIRAL SCAN STUDY CYCLE TIME (30 Spiral Slice Sequence)

Time from first to last exposure =	_____ (≤ 48 sec.)
Total time from first exposure to last image review =	_____ (≤ 80 sec.)

TEST 11: BACKUP TIMER (OPTIONAL)

TIME SPECIFICATIONS

Description	Results Obtained	Specification	Initials	Date
4.2 sec. Scan time	sec	3.79 to 4.61 secs.	NA	NA
2 sec. Scan time	sec	1.80 to 2.19 secs.	NA	NA
Full field pilot	sec	4.20 to 6.00 secs.	NA	NA
Back-up timer	sec	≤ 4.45 secs.	NA	NA
Back-up timer	sec	≤ 2.225 secs	NA	NA